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資料結構

Data Structure

Lab 06

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Lab06-Ex1.
玩 Blackjack
Code

```
#include <iostream>
#include <vector>
#include <string>
#include <algorithm>
#include <random>
#include <ctime>
using namespace std;

struct Player {// 玩家結構體，包含名稱和分數
    string name; // 玩家名稱
    int score; // 玩家分數
};

class Card {//建立一個 Card 類別，用來表示單張撲克牌
public:
    string suit; // 花色
    string rank; // 點數
    Card() {}
    Card(string s, string r) : suit(s), rank(r) {}
    void display() const {
        cout << rank << " of " << suit << endl;
    }
};

class Deck {// 牌組類別，負責生成和洗牌
private:
    Card cards[52]; // 52 張牌
    int index; // 目前發牌索引
public:
    Deck() : index(0) {// 初始化牌組
        string suits[] = { "Hearts", "Diamonds", "Clubs", "Spades" };// 四種花色
        string ranks[] = { "2", "3", "4", "5", "6", "7", "8", "9", "10", "J",
"Q", "K", "A" };// 13 張牌
        int k = 0;
```

```
        for (int i = 0; i < 4; i++) { // 4 種花色
            for (int j = 0; j < 13; j++) { // 13 種點數
                cards[k] = Card(suits[i], ranks[j]);
                k++;
            }
        }
    }

    void shuffleDeck() {// 洗牌
        srand(time(0));
        for (int i = 51; i > 0; i--) {// 從最後一張牌開始，向前遍歷
            int j = rand() % (i + 1);// 生成隨機索引 j
            swap(cards[i], cards[j]);// 交換牌組中的兩張牌
        }
    }

    // 取得指定位置的牌
    Card getCard(int i) const {
        return cards[i];
    }
};

// Queue 類別
class Queue {
private:
    vector<Card> deck; // 用來存放卡牌的容器
    int Front; // 指向牌堆的前端
    int Rear; // 指向牌堆的尾端
    int capacity; // 容器最大容量
public:
    // 建構函式
    Queue(int size) : Front(-1), Rear(-1), capacity(size) {}

    // 插入元素到佇列尾端
    void enqueue(Card card) {
        if (isFull()) { // 檢查是否滿
            cout << "牌堆已滿，無法插入！" << endl;
            return;
```

```
        }
        if (Front == -1) {
        Front = 0;
        }
        deck.push_back(card); // 插入元素到尾端
        Rear++; // 尾端指標移動
    }

    // 從佇列中移除最前端元素
    void dequeue() {
        if (isEmpty()) {//檢查是不是空的
            cout << "牌堆為空，無法移除！" << endl;
            return;
        }
        Front++;//把 front 往後移一格
        if (Front > Rear) {
            Front = -1;//把 front 重設
            Rear = -1;//把 rear 重設
            deck.clear();//把 deck 清空
        }
    }

    Card front() {// 獲取佇列最前端元素
        if (isEmpty()) { // 檢查是否空
            cout << "牌堆為空，無法獲取最前端的牌！" << endl;
            return Card("", "");
        }
        return deck[Front];
    }

    bool isEmpty() const {// 判斷佇列是否為空
        return (Front == -1 && Rear == -1) || (Front > Rear && Front != -1);//判
斷是不是空的
    }

    bool isFull() const {// 判斷佇列是否已滿
        return deck.size() == capacity;// 判斷是不是滿的
    }
```

```
};

void initializeDeck(Queue& cardDeck) {// 初始化並洗牌
    Deck deck;
    cout << "初始牌堆: \n";
    for (int i = 0; i < 52; i++) {
        deck.getCard(i).display(); //印出原始牌堆
    }
    deck.shuffleDeck();//洗牌
    cout << "\n 洗牌後的牌堆: \n";
    for (int i = 0; i < 52; i++) {
        deck.getCard(i).display();//印出洗牌後的牌堆
    }
    for (int i = 0; i < 52; i++) { //
    cardDeck.enqueue(deck.getCard(i)); // //把牌放進 queue
    }
}

void initializePlayer(Player* player, string name, Queue& cardDeck)
{// 初始化玩家，發兩張牌
    player->name = name;// 玩家名稱
    player->score = 0;// 玩家分數
    cout << player->name << " 抽到的牌: ";
    for (int i = 0; i < 2; i++) {// 發兩張牌
        Card drawnCard = cardDeck.front();//看最前面的牌是什麼
        cardDeck.dequeue();//把最前面的牌移除
        // 將抽到的牌印出
        cout << drawnCard.rank << " of " << drawnCard.suit << " ";
        if (drawnCard.rank == "A") player->score += 1;//計算 A 分數
        else if (drawnCard.rank == "J" || drawnCard.rank == "Q" || drawnCard.rank
== "K") player->score += 10;//計算 JQK 分數
        else player->score += stoi(drawnCard.rank);//計算 2-10 分數
    }
    cout << "\n";
    cout << player->name << " 玩家的初始分數: " << player->score << endl;//印出玩家
的初始分數
}
```

```
void playerTurn(Player* player, Queue& cardDeck) {// 玩家回合
    char choice;
    while (player->score < 21 && !cardDeck.isEmpty()) {// 玩家小於 21 點且牌堆不為
空
        cout << player->name << " 您的手牌分數目前為: " << player->score << " 要抽
牌嗎？(h = 抽, s = 停) : ";//顯示選擇
        cin >> choice;//輸入選擇
        if (choice == 'h') {//如果玩家選擇抽牌
            Card newCard = cardDeck.front();
            cardDeck.dequeue();
            if (newCard.rank == "A") player->score += 1;//計算 A 點數
            else if (newCard.rank == "J" || newCard.rank == "Q" || newCard.rank
== "K") player->score += 10; //計算 JQK 點數
            else player->score += stoi(newCard.rank);//計算 2-10 點數
            //印出抽到的牌和總點數
            cout << player->name << " 抽到: " << newCard.rank << " of " <<
newCard.suit << " 總分: " << player->score << endl;
            if (player->score > 21) {//如果玩家大於 21 點
                cout << player->name << " 爆掉了！遊戲結束。\n";//玩家爆牌，莊家獲勝
                return;
            }
        }
        else if (choice == 's') {//如果玩家選擇停牌
            cout << player->name << " 選擇停牌，總分: " << player->score << endl;//
印出總點數
            break;
        }
        else {
            cout << "請輸入有效選項 (h = 抽, s = 停)！" << endl;//輸入無效
        }
    }
}

void dealerTurn(Player* dealer, Queue& cardDeck) {// 莊家回合
    while (dealer->score < 17 && !cardDeck.isEmpty()) {// 莊家小於 17 點且牌堆不為
空
        Card newCard = cardDeck.front();
        cardDeck.dequeue();
```

```
        if (newCard.rank == "A") dealer->score += 1;//計算 A 點數
        else if (newCard.rank == "J" || newCard.rank == "Q" || newCard.rank ==
"K") dealer->score += 10;//計算 JQK 點數
        else dealer->score += stoi(newCard.rank);//計算 2-10 點數
        //印出抽到的牌和總點數
        cout << "莊家抽到: " << newCard.rank << " of " << newCard.suit << " 莊家目
前總分: " << dealer->score << endl;
        if (dealer->score > 21) {//如果莊家大於 21 點
            cout << "莊家爆了！玩家獲勝！" << endl;//玩家獲勝
            return;
        }
    }
}

// 判斷勝負
void determineWinner(Player* player, Player* dealer) {
    if (player->score > 21) cout << player->name << " 爆了！莊家獲勝！\n";//玩家爆
牌，莊家獲勝
    else if (dealer->score > 21 || player->score > dealer->score)
        cout << player->name << " 贏了！\n";//莊家爆牌或玩家點數大於莊家，玩家獲勝
    else if (player->score == dealer->score) cout << "平手！\n";//莊家與玩家點數相
同，平手
    else cout << "莊家贏了！\n";
}

int main() {
    srand(time(0));
    Queue cardDeck(52);//宣告牌堆
    initializeDeck(cardDeck);//初始化牌堆
    Player player, dealer;//宣告並初始化莊家以及玩家
    initializePlayer(&player, "玩家", cardDeck);
    initializePlayer(&dealer, "莊家", cardDeck);
    playerTurn(&player, cardDeck);
    if (player.score <= 21) {
        cout << "\n 莊家回合...\n";
        dealerTurn(&dealer, cardDeck);
        determineWinner(&player, &dealer);
    }
```

```
    return 0;
}
```

Result

C:\Users\User\Desktop\資料結構作業\Lab06\Lab06_Ex1.exe

```
初始牌堆:
2 of Hearts
3 of Hearts
4 of Hearts
5 of Hearts
6 of Hearts
7 of Hearts
8 of Hearts
9 of Hearts
10 of Hearts
J of Hearts
Q of Hearts
K of Hearts
A of Hearts
2 of Diamonds
3 of Diamonds
4 of Diamonds
5 of Diamonds
6 of Diamonds
7 of Diamonds
8 of Diamonds
9 of Diamonds
10 of Diamonds
J of Diamonds
Q of Diamonds
K of Diamonds
A of Diamonds
2 of Clubs
3 of Clubs
4 of Clubs
5 of Clubs
6 of Clubs
7 of Clubs
8 of Clubs
9 of Clubs
10 of Clubs
J of Clubs
Q of Clubs
K of Clubs
A of Clubs
2 of Spades
3 of Spades
4 of Spades
5 of Spades
6 of Spades
7 of Spades
8 of Spades
9 of Spades
10 of Spades
```

C:\Users\User\Desktop\資料結構作業\Lab06\Lab06_Ex1.exe

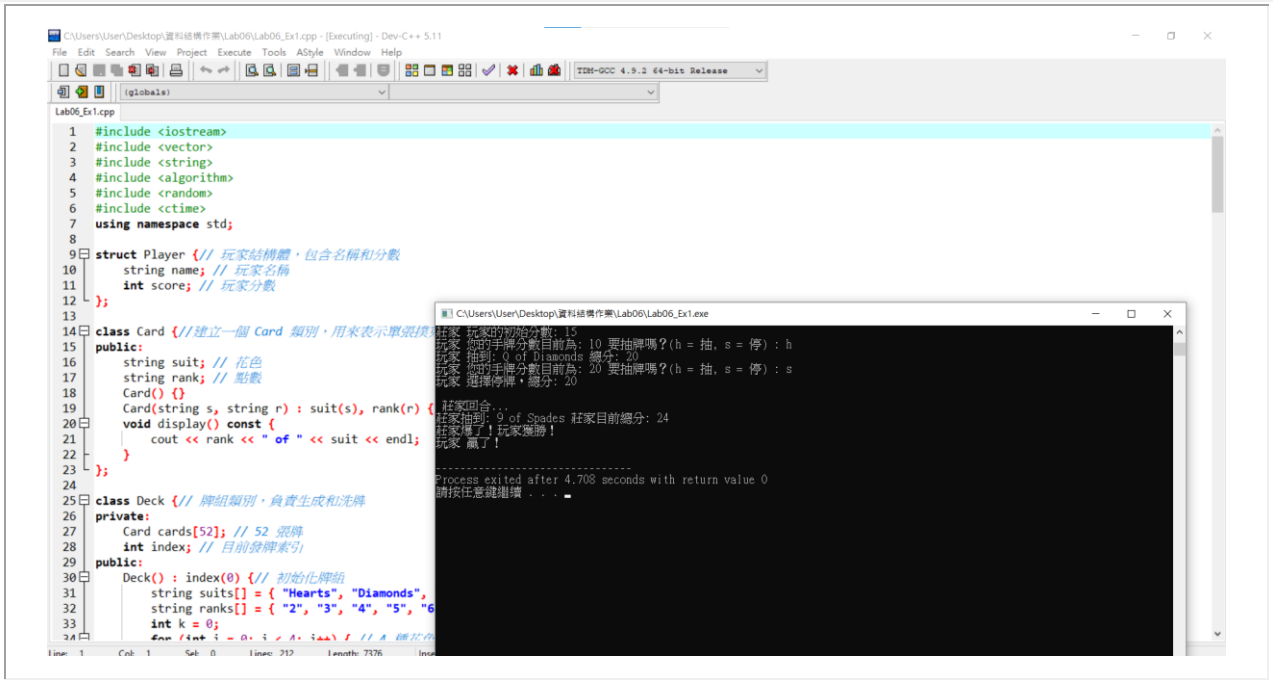
```
初始牌堆:
2 of Hearts
3 of Hearts
4 of Hearts
5 of Hearts
6 of Hearts
7 of Hearts
8 of Hearts
9 of Hearts
10 of Hearts
J of Hearts
Q of Hearts
K of Hearts
A of Hearts
2 of Diamonds
3 of Diamonds
4 of Diamonds
5 of Diamonds
6 of Diamonds
7 of Diamonds
8 of Diamonds
9 of Diamonds
10 of Diamonds
J of Diamonds
Q of Diamonds
K of Diamonds
A of Diamonds
2 of Clubs
3 of Clubs
4 of Clubs
5 of Clubs
6 of Clubs
7 of Clubs
8 of Clubs
9 of Clubs
10 of Clubs
J of Clubs
Q of Clubs
K of Clubs
A of Clubs
2 of Spades
3 of Spades
4 of Spades
5 of Spades
6 of Spades
7 of Spades
8 of Spades
9 of Spades
10 of Spades
```

C:\Users\User\Desktop\資料結構作業\Lab06\Lab06_Ex1.exe

```
K of Clubs
8 of Clubs
Q of Clubs
Q of Hearts
3 of Spades
2 of Diamonds
10 of Diamonds
7 of Hearts
8 of Diamonds
A of Spades
8 of Hearts
3 of Clubs
2 of Spades
J of Hearts
K of Spades
K of Diamonds
8 of Spades
玩家 抽到的牌: 3 of Hearts 7 of Spades
玩家 玩家的初始分數: 10
莊家 抽到的牌: 10 of Spades 5 of Spades
莊家 玩家的初始分數: 15
玩家 您的手牌分數目前為: 10 要抽牌嗎?(h = 抽, s = 停) : h
玩家 抽到: Q of Diamonds 總分: 20
玩家 您的手牌分數目前為: 20 要抽牌嗎?(h = 抽, s = 停) : s
玩家 選擇停牌，總分: 20

 莊家回合...
莊家抽到: 9 of Spades 莊家目前總分: 24
莊家爆了！玩家獲勝！
玩家 贏了！

--------------------------------
Process exited after 4.708 seconds with return value 0
請按任意鍵繼續 . . .
```



Discussion

Stack 是最後進去的先出去，

Queue 是最先進來的先出去